



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Chemical Engineering
In-Semester Examination / Evaluation -II
Academic Year: 2025-26 (semester I)

Class: SY

Course: Fluid Mechanics

Course Code: PCC-CH201/202

Date: 14.10.2025

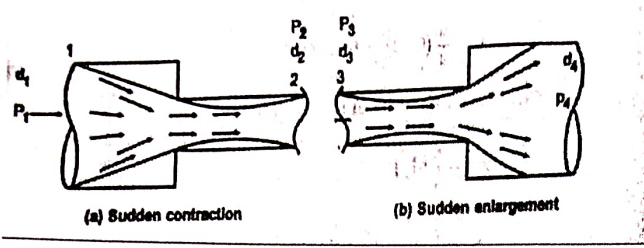
Time: 14.30 to 15:15

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- All questions are compulsory
- Figures to right indicate full marks.
- The symbols carry usual meaning.
- This paper contains 02 pages

Q.N.			Marks	CO	BT
Q.1	a	A contractor at SGGSI&T Nanded designed and laid the pipeline from the Panchal Canteen to Ten-Bhagyanagar Canteen. The viscosity of water is 1 poise and relative density is 1. The water circulation line is 20m long with pipe diameter 20mm. The datum head at Ten-Bhagyanagar Canteen is 20m. The pressure head at Panchal Canteen is read 6 kgf/m² and at Bhagyanagar Canteen is 2 kgf/m². Analyze, inspect and propose whether the flow will be occurred from the Panchal Canteen to Bhagyanagar Canteen? Also evaluate the mass flow rate of water available at Ten-Bhagyanagar Canteen. (Consider frictional loss only)	05	CO3	BT6
	b	In the soap manufacturing plant the reactor inlet line carries the fluid of density 1000kg/m³. The diameter of inlet pipeline is 5cm. The volumetric flow rate of the fluid is 0.003m³/s and length of the pipe line is 400m. Evaluate the following - Average velocity of fluid - maximum velocity of fluid - The velocity of the fluid at a distance 1.5 cm from the pipe wall - Head lost due to friction - Pressure drop across the pipe - Shear force at pipe wall Viscosity of fluid is 0.1N-S/m²	05	CO3	BT5
Q.2	a	As shown in fig. below when a sudden contraction is introduced in a horizontal pipeline from 50cm to 25cm, the pressure changes from 10500kgf/cm² to 6900kgf/cm². Calculate the rate of flow. Assume coefficient of contraction of jet to be 0.65. Following this if there is a sudden enlargement from 25cm to 50cm and if the pressure at the 25cm section is 6900kgf/m² what is the pressure at the 50cm enlarged section.	05	CO4	BT4

		 (a) Sudden contraction (b) Sudden enlargement			
	b	Initiating from Newton's Second Law of Motion systematically demonstrate the development of the following fundamental equations of fluid motion i) Reynolds Equation of Motion ii) Navier–Stokes Equation iii) Euler's Equation of Motion Further, from Euler's Equation of motion, derive the Bernoulli's Equation for steady, incompressible, and inviscid flow, clearly stating all necessary assumptions and physical interpretations at each step. Also Explain Bernoulli's Principle	05	CO4	BT3

=== Think, Apply, and Succeed. ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	To understand the basics rheology of the fluids and fluid properties
CO 2	To understand the concept of friction factor and analyze the fluid flow phenomena in various regions
CO 3	To do the mass, momentum and energy balance and will be able to understand working of various flow meters.
CO 4	To understand and analyze and design the various flow-meters
CO 5	To understand and analyze various fluid transportation pumps
CO6	To understand and various types of pipe fitting, pumps and their performance characteristics.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Chemical Engineering
In-Semester Examination II
Academic Year: 2025-26 (Semester-I)

Class: Second Year	Subject: Chemical Process Calculations	Subject Code: MDM-CH201
Date: 16.10.2025	Time: 14.30 to 15:15	Max. Marks: 20

Note:

- i. Attempt any four questions
- ii. Assume suitable data whenever necessary and state it clearly

Q.N.	Question	Marks	CO	BT
Q.1	Define the following terms: i. Limiting reactant iii. Recycle ratio v. Purge ratio ii. Conversion iv. Yield	05	2	1
Q.2	In the production of sulphur trioxide, 100 kmol of SO ₂ and 100 kmol of O ₂ are fed to a reactor. If the percent conversion of SO ₂ is 80, calculate the composition of the product stream on a mole basis.	05	3	5
Q.3	The following reaction describes the oxidation of ethylene to produce ethylene oxide. $C_2H_4 + \frac{1}{2} O_2 \rightarrow C_2H_4O$ If air is used in 20% excess of that theoretically required, calculate the quantity of air supplied based on 100 kmol of ethylene fed to the reactor.	05	3	5
Q.4	Formaldehyde is produced by the dehydrogenation of methanol. $CH_3OH \rightarrow HCHO + H_2$ The per-pass conversion is 67%. The product leaving the reactor is fed to a separation unit battery where formaldehyde is separated from methanol and hydrogen. The separated methanol is recycled to the reactor. If the production rate of formaldehyde is 1000 kg/h. Calculate: (a) the combined feed ratio (b) the flow rate of methanol required to the process (as fresh feed) and (c) purge ratio	05	3	5
Q.5	Explain the need for recycle and purging operations in chemical process industries in detail.	05	4	4

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1	Know the basic concepts regarding unit systems and conversions in the chemical engineering process
CO2	Able to carry out the material balancing in chemical processes without chemical reactions
CO3	Able to carry out the material balancing in chemical processes involving chemical reactions
CO4	Able to carry out the energy balancing in chemical process involving chemical reactions and without chemical reactions
CO5	Able to provide solution to various unit operations.



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Department of Chemical Engineering
In-Semester Examination II
Academic Year: 2025-26 (semester I)

Class: Second Year B. Tech.

Course: Environmental Chemistry and Microbiology

Course Code:
MDM-IS-CH201

Date: 16-Oct-2025

Time: 14:30 to 15:15

Max. Marks: 20

Note:

- All questions are compulsory.*
- Assume suitable data whenever necessary and state it clearly.*
- Figures to right indicate full marks.*
- The symbols carry usual meaning.*

Q.N.			Marks	CO	BT
Q.1	a	Write a short note on the role of agricultural chemistry?	4	CO2	BT-1
	b	What is cation exchange capacity, and how does it function in soil chemistry?	4	CO2	BT-1
	c	What is the classification of microorganisms? Write about four major types of microbes?	4	CO3	BT-1
Q.2	a	What is the transmission of pathogens? Explain the different types of disease transmission?	4	CO3	BT-1
	b	Define the distribution of microorganisms in different environments?	4	CO3	BT-1

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand the basics of stoichiometry and mass balance of reactions related to chemical and environmental science
CO 2	Understand the basic of chemical and photochemical reaction and nature and composition of soil
CO 3	Understand the classification, distribution, diversity and importance of microorganism in different environments
CO 4	Understand the Nutrition and Metabolism in microorganism and growth phases
CO 5	Understand the fundamental of Ecotoxicology toxicants and toxicity



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Department of Chemical Engineering
In-Semester Examination II
Academic Year: 2025-26 (semester I)

Class: Second Year B. Tech.

Course: Environmental Law

Course Code:
OE- IS-CH201

Date: 17-Oct-2025

Time: 14:30 to 15:15

Max. Marks: 20

Note:

- i. *All questions are compulsory.*
- ii. *Assume suitable data whenever necessary and state it clearly.*
- iii. *Figures to right indicate full marks.*

Q.N.			Marks	CO	BT
Q.1	a	What is polluter pays principle? Explain the land mark judgement on Polluter Pays Principle ("PPP") by Supreme Court in the case of Indian Council for Enviro Legal Action Vs Union of India on Bichhari Village of Rajasthan	4	CO2	BT-1
	b	Explain the Doctrine of Absolute Liability with Different Case Laws?	4	CO2	BT-3
Q.2	a	What is noise pollution, and what are the major causes of noise pollution?	4	CO3	BT-I
	b	Write a short note on the effects of noise pollution. Mention at least five major problems caused by it?	4	CO3	BT-I
	c	What are the restrictions on the dereservation of forests or the use of forest land for non-forest purposes?	4	CO4	BT-I

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Explain the fundamental principle of environmental law, including sustainable development, precautionary principle, and polluter pays principle.
CO 2	Interpret constitutional provision, national legislation, and international conventions related to environment protection.
CO 3	Apply legal frameworks to analyse case studies on water pollution, air pollution, and hazardous waste management
CO 4	Examine landmark judicial decision (e.g. M.C. Mehta cases) and evaluate their impact on environmental governance in India.
CO 5	Assess the role of regulatory bodies, policies, and public participation in effective environment management.
CO 6	Develop solution and recommendation for industrial and community level environmental issue by integrating legal and technical knowledge.